

CS6360-Assignment 3

Proof Presentation

*Compositional Learning of Dynamical System Models Using Port
Hamiltonian Neural Networks*

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Port-Hamiltonian Systems Model

where $\dot{x} = [J(x) - R(x)] \nabla_x H(x) + G(x)u, \quad y = G(x)^T \nabla_x H(x)$

- $x \in R^n$ is n dimensional vector representing the system's state
- $H(x)$ is the system's Hamiltonian function
- $J(x) \in R^{n \times n}$ is the skew symmetric interconnection matrix (i.e. $J(x) = -J(x)^T$ for every $x \in R^n$)
- $R(x) \in R^{n \times n}$ is the symmetric positive semi definite dissipation matrix
- $G(x) \in R^{n \times m}$ is control input matrix
- $u \in R^m$ is m dimensional control input vector and $y \in R^m$ is output vector

$H(x)$ describes the system's energy in terms of its state x , $J(x)$ encodes the energy-conserving interactions between the various components of the system and $R(x)$ encodes how these components dissipate energy.

Background

Composite Model (Terms): x : system's state, u : control input

- $P_i(x_i, u_i)$: True Subsystem dynamics
- $P_{i,\theta}(x_i, u_i)$: Parameterized subsystem dynamics
- $P_c(x_c, u_c)$: Composite system
- $C(x_c)$: Composition term
- $C_\phi(x_c)$: Parameterized composition term
- $C_\phi(x_c)_{i,j}$: Interaction between i and j

For every $i, j = 1, \dots, n$ we have

$$\begin{aligned} \left\| P_i(x_i, u_i) - P_{i,\theta}(x_i, u_i) \right\| &\leq \epsilon_i \\ \left\| \nabla_{x_i} H_i(x_i) - \nabla_{x_i} H_{i,\theta}(x_i) \right\| &\leq \eta_i \end{aligned}$$

Also,

$$\left\| (C(x_c)_{i,j} - C_\phi(x_c)_{i,j}) \nabla_{x_j} H_j(x_j) \right\| \leq \gamma_{i,j}$$

Bounding Composition Errors of PHNNs

Theorem 1 *Suppose the true dynamics of each subsystem may be written in port-Hamiltonian form as $P_i(\mathbf{x}_i, \mathbf{u}_i)$ for $i = 1, 2, \dots, k$. Furthermore, suppose the composite system of interest may be represented as a composition $P_c^C(\mathbf{x}_c, \mathbf{u}_c)$ of the port-Hamiltonian subsystems defined by the composition term $C(\mathbf{x}_c)$. Let $C_\phi(\mathbf{x}_c)$ denote the learned composition term and let $C_\phi(\mathbf{x}_c)_{i,j}$ denote the submatrix that defines the interactions between subsystems i and j . Suppose that for every $i, j = 1, \dots, k$ with $i \neq j$, we have $\|P_i(\mathbf{x}_i, \mathbf{u}_i) - P_{i,\Theta}(\mathbf{x}_i, \mathbf{u}_i)\| \leq \varepsilon_i$ and $\|\nabla_{\mathbf{x}_i} H_i(\mathbf{x}_i) - \nabla_{\mathbf{x}_i} H_{i,\theta}(\mathbf{x}_i)\| \leq \eta_i$ for all $\mathbf{x}_i \in \Omega_{\mathbf{x}_i}$, $\mathbf{u}_i \in \Omega_{\mathbf{u}_i}$. Also suppose that $\|(C(\mathbf{x}_c)_{i,j} - C_\phi(\mathbf{x}_c)_{i,j})\nabla_{\mathbf{x}_j} H_j(\mathbf{x}_j)\| \leq \gamma_{i,j}$ for all $\mathbf{x}_j \in \Omega_{\mathbf{x}_j}$ that are consistent with some composite state $\mathbf{x}_c \in \Omega_{\mathbf{x}_1} \times \dots \times \Omega_{\mathbf{x}_k}$. Then,*

$$\left\| P_c^C(\mathbf{x}_c, \mathbf{u}_c) - P_{c,\Theta_c}^{C_\phi}(\mathbf{x}_c, \mathbf{u}_c) \right\| \leq \sum_{i=1}^k [\varepsilon_i + 2 \sum_{j>i}^k (\gamma_{i,j} + \sigma_{i,j} \eta_j)], \quad (5)$$

for every $\mathbf{x}_c \in \Omega_{\mathbf{x}_1} \times \dots \times \Omega_{\mathbf{x}_k}$ and $\mathbf{u}_c \in \Omega_{\mathbf{u}_1} \times \dots \times \Omega_{\mathbf{u}_k}$. Here, $\sigma_{i,j} := \max_{\mathbf{x}_c \in \Omega_{\mathbf{x}_c}} \|C_\phi(\mathbf{x}_c)_{i,j}\|$ where $\|C_\phi(\mathbf{x}_c)_{i,j}\| := \sup\{\|C_\phi(\mathbf{x}_c)_{i,j}y\| \text{ s.t. } y \in \mathbb{R}^{n_j}, \|y\| = 1\}$ is the matrix norm of $C_\phi(\mathbf{x}_c)_{i,j}$.

Theorem

- It tells us that the ***prediction error*** of the composite PHNN is **bounded** by the error introduced by its *component models* and the error introduced by the *influence* that its component have on each other.
- Function of error in our estimate of the composition term $C_\phi(x)$ and in the gradients of the subsystem Hamiltonian functions $\nabla_{x_i} H_{i,\theta}(x_i)$

$$\left\| P_c^C(\mathbf{x}_c, \mathbf{u}_c) - P_{c, \Theta_c}^{C_\phi}(\mathbf{x}_c, \mathbf{u}_c) \right\| \leq \sum_{i=1}^k [\varepsilon_i + 2 \sum_{j>i}^k (\gamma_{i,j} + \sigma_{i,j} \eta_j)],$$

Proof

- Expand the expression for the error in the composite PHNN.

$$\|P_c(\mathbf{x}_c, \mathbf{u}_c) - P_{c, \Theta_c}^{C_\phi}(\mathbf{x}_c, \mathbf{u}_c)\| \quad (7)$$

$$= \left\| \left[\tilde{\mathbf{J}}_c(\mathbf{x}_c) + \mathbf{C}(\mathbf{x}_c) - \mathbf{R}_c(\mathbf{x}_c) \right] \nabla_{\mathbf{x}_c} H_c(\mathbf{x}_c) + \mathbf{G}_c(\mathbf{x}_c) \mathbf{u}_c \right. \\ \left. - \left[\tilde{\mathbf{J}}_c(\mathbf{x}_c) + \mathbf{C}_\phi(\mathbf{x}_c) + \mathbf{R}_{c, \theta}(\mathbf{x}_c) \right] \nabla_{\mathbf{x}_c} H_{c, \theta}(\mathbf{x}_c) + \mathbf{G}_{c, \theta}(\mathbf{x}_c) \mathbf{u}_c \right\| \quad (8)$$

$$\leq \left\| \left[\tilde{\mathbf{J}}_c(\mathbf{x}_c) - \mathbf{R}_c(\mathbf{x}_c) \right] \nabla_{\mathbf{x}_c} H_c(\mathbf{x}_c) + \mathbf{G}_c(\mathbf{x}_c) \mathbf{u}_c \right. \\ \left. - \left[\tilde{\mathbf{J}}_c(\mathbf{x}_c) - \mathbf{R}_{c, \theta}(\mathbf{x}_c) \right] \nabla_{\mathbf{x}_c} H_{c, \theta}(\mathbf{x}_c) - \mathbf{G}_{c, \theta}(\mathbf{x}_c) \mathbf{u}_c \right\| \\ + \left\| \mathbf{C}(\mathbf{x}_c) \nabla_{\mathbf{x}_c} H_c(\mathbf{x}_c) - \mathbf{C}_\phi(\mathbf{x}_c) \nabla_{\mathbf{x}_c} H_{c, \theta}(\mathbf{x}_c) \right\| \quad (9)$$

$$\leq \sum_{i=1}^k \left\{ \left\| \left[\mathbf{J}_i(\mathbf{x}_i) - \mathbf{R}_i(\mathbf{x}_i) \right] (\nabla_{\mathbf{x}_c} H_c(\mathbf{x}_c))_{\mathcal{I}_i} + \mathbf{G}_i(\mathbf{x}_i) \mathbf{u}_i \right. \right. \\ \left. \left. - \left[\mathbf{J}_{i, \theta}(\mathbf{x}_i) - \mathbf{R}_{i, \theta}(\mathbf{x}_i) \right] (\nabla_{\mathbf{x}_c} H_{c, \theta}(\mathbf{x}_c))_{\mathcal{I}_i} + \mathbf{G}_{i, \theta}(\mathbf{x}_i) \mathbf{u}_i \right\| \right\} \\ + \left\| \mathbf{C}(\mathbf{x}_c) \nabla_{\mathbf{x}_c} H_c(\mathbf{x}_c) - \mathbf{C}_\phi(\mathbf{x}_c) \nabla_{\mathbf{x}_c} H_{c, \theta}(\mathbf{x}_c) \right\| \quad (10)$$

Proof

- Define $J_c(x_c) := \text{Diag}(J_1(x_1), \dots, J_k(x_k))$.
- Equation 9 follows from equation 8 by the triangle inequality.
- Equation 10 follows from equation 9 again by the triangle inequality and due to the block-diagonal structure of $J_c(x_c)$, $R_c(x_c)$, and $G_c(x_c)$.
- In equation 10 we use I_i to denote the set of indexes corresponding to subsystem i within the composite vector x_c .
- $\left(\nabla_{x_c} H_c(x_c)\right)_{I_i}$ denote the vector in R^{n_i} that results from taking the elements from $\nabla_{x_c} H_c(x_c)$ indexed by I_i and discarding the rest of the vector. Then, by definition of $H_{x_c}(x_c)$, $\nabla_{x_c} H_c(x_c) = \sum_{j=1}^k \nabla_{x_c} H_j(x_j)$.

Proof

- Equation 11 is given below

$\nabla_{\mathbf{x}_c} H_c(\mathbf{x}_c) = \sum_{j=1}^k \nabla_{\mathbf{x}_c} H_j(\mathbf{x}_j)$. For every $i \neq j$, we have that $(\nabla_{\mathbf{x}_c} H_j(\mathbf{x}_j))_{\mathcal{I}_i} = \mathbf{0} \in \mathbb{R}^{n_i}$ because of our assumption that the subsystem Hamiltonian $H_j(\cdot)$ only depends on $\mathbf{x}_j$. Furthermore, using the same reasoning it must be true that $(\nabla_{\mathbf{x}_c} H_j(\mathbf{x}_j))_{\mathcal{I}_j} = \nabla_{\mathbf{x}_j} H_j(\mathbf{x}_j)$. So, we conclude that

$$(\nabla_{\mathbf{x}_c} H_c(\mathbf{x}_c))_{\mathcal{I}_i} = \sum_{j=1}^k (\nabla_{\mathbf{x}_c} H_j(\mathbf{x}_j))_{\mathcal{I}_i} = \nabla_{\mathbf{x}_i} H_i(\mathbf{x}_i). \quad (11)$$

Proof

- Combining equation 11 with equation 10, we have

$$\|P_c(\mathbf{x}_c, \mathbf{u}_c) - P_{c, \Theta_c}^{C_\phi}(\mathbf{x}_c, \mathbf{u}_c)\| \leq \sum_{i=1}^k \|P_i(\mathbf{x}_i, \mathbf{u}_i) - P_{i, \Theta}(\mathbf{x}_i, \mathbf{u}_i)\| \quad (12)$$

$$\begin{aligned} &+ \|C(\mathbf{x}_c) \nabla_{\mathbf{x}_c} H_c(\mathbf{x}_c) - C_\phi(\mathbf{x}_c) \nabla_{\mathbf{x}_c} H_{c, \theta}(\mathbf{x}_c)\| \\ &\leq \sum_{i=1}^k \varepsilon_i + \|C(\mathbf{x}_c) \nabla_{\mathbf{x}_c} H_c(\mathbf{x}_c) - C_\phi(\mathbf{x}_c) \nabla_{\mathbf{x}_c} H_{c, \theta}(\mathbf{x}_c)\|. \quad (13) \end{aligned}$$

Proof

Now, bound the final term in the right-hand side of equation 13

$$\|C(\mathbf{x}_c)\nabla_{\mathbf{x}_c}H_c(\mathbf{x}_c) - C_\phi(\mathbf{x}_c)\nabla_{\mathbf{x}_c}H_{c,\theta}(\mathbf{x}_c)\| \quad (14)$$

$$\leq \| (C(\mathbf{x}_c) - C_\phi(\mathbf{x}_c))\nabla_{\mathbf{x}_c}H_c(\mathbf{x}_c) \| + \| C_\phi(\mathbf{x}_c)(\nabla_{\mathbf{x}_c}H_c(\mathbf{x}_c) - \nabla_{\mathbf{x}_c}H_{c,\theta}(\mathbf{x}_c)) \| \quad (15)$$

$$\leq \sum_{\substack{i=1 \\ j \neq i}}^k \{ \| (C(\mathbf{x}_c)_{\mathcal{I}_i, \mathcal{I}_j} - C_\phi(\mathbf{x}_c)_{\mathcal{I}_i, \mathcal{I}_j})\nabla_{\mathbf{x}_j}H_j(\mathbf{x}_j) \| \quad (16)$$

$$+ \| C_\phi(\mathbf{x}_c)_{\mathcal{I}_i, \mathcal{I}_j}(\nabla_{\mathbf{x}_j}H_j(\mathbf{x}_j) - \nabla_{\mathbf{x}_j}H_{j,\theta}(\mathbf{x}_j)) \| \} \quad (17)$$

$$\leq 2 \sum_{\substack{i=1 \\ j > i}}^k \{ \gamma_{i,j} + \| C_\phi(\mathbf{x}_c) \| \| \nabla_{\mathbf{x}_j}H_j(\mathbf{x}_j) - \nabla_{\mathbf{x}_j}H_{j,\theta}(\mathbf{x}_j) \| \} \quad (18)$$

$$\leq 2 \sum_{\substack{i=1 \\ j > i}}^k \gamma_{i,j} + \sigma_{i,j}\eta_j \quad (19)$$

Proof

- Equation 15 follows from equation 14 by adding and subtracting $C_\phi(x_c)$, $\nabla_{x_c} H_c(x_c)$, regrouping terms, and applying the triangle inequality.
- Obtain equation 16 by rewriting the matrix multiplications from equation 15 using the submatrices $C(x_c)_{I_i, I_j}$ of $C(x_c)$ (which correspond to the interactions between subsystem i and j).
- Exclude $C(x_c)_{I_i, I_i}$ from the sum in equation 16 because, by definition, $C(x_c)_{I_i, I_i} = 0$ for all $i = 1, 2, \dots, k$.
- Combining equation 19 with equation 13, gives the desired result.

Tools

- Dirac Structures
- Components of PHS

Dirac Structure

- Definition

Definition 2.1. Consider a finite-dimensional linear space $\mathcal{F}$ with $\mathcal{E} = \mathcal{F}^*$. A subspace $\mathcal{D} \subset \mathcal{F} \times \mathcal{E}$ is a *Dirac structure* if

1. $\langle e | f \rangle = 0$, for all $(f, e) \in \mathcal{D}$,
2. $\dim \mathcal{D} = \dim \mathcal{F}$.

- Property (1) corresponds to **power-conservation**, and expresses the fact that the total power entering (or leaving) a Dirac structure is zero.
- Maximal dimension of any subspace $\mathcal{D} \subset \mathcal{F} \times \mathcal{E}$ satisfying (1) = $\dim \mathcal{F}$.

PHS Theory

- Energy Storing Elements
- Energy dissipating Elements
- External (Interaction) Ports
- Port Hamiltonian Dynamics
- Power balance

Energy Storing Elements

- Multi-port element S corresponds to the union of all the energy-storing elements of the system.
- Port variable product $(e_S^T f_S)$ denoting the total power flowing into the Dirac structure from the energy storing elements.
- Power-balance for the energy-storing multi-port element

$$\frac{d}{dt}H = \left\langle \frac{\partial H}{\partial x}(x) \mid \dot{x} \right\rangle = \frac{\partial^T H}{\partial x}(x)\dot{x}.$$

- Interconnection of the energy-storing elements to the storage port (e_S, f_S) of the Dirac structure is accomplished by setting

$$f_S = -\dot{x} \quad \text{and} \quad e_S = \frac{\partial H}{\partial x}(x).$$

- Power-balance can be also written as

$$\frac{d}{dt}H = \frac{\partial^T H}{\partial x}(x)\dot{x} = -e_S^T f_S.$$

Energy Dissipating elements

- Multi-port element R corresponds to internal energy dissipation (due to friction, resistance, etc.), and its port variables are denoted by (e_R, f_R) .
- Property

$$\langle e_R | f_R \rangle = e_R^T f_R \leq 0$$

- For all the subset R an energy-dissipating relation, satisfy the power balance

$$e_S^T f_S + e_R^T f_R = 0$$

- Power balance relation is given by

$$\frac{d}{dt} H = -e_S^T f_S = e_R^T f_R \leq 0$$

External Ports

- External port (f_P, e_P) models the interaction of the system with its environment
- Types: Controller port variables and Interaction port variables (Source)
- Power balance with external port is given by

$$e_S^T f_S + e_R^T f_R + e_P^T f_P = 0$$

- Extends energy dissipation equation to

$$\frac{d}{dt} H = e_R^T f_R + e_P^T f_P \leq e_P^T f_P, \quad e_R^T f_R \leq 0.$$

- This inequality expresses that the increase of the internally stored energy (the Hamiltonian) is always less than or equal to the externally supplied power.

Port Hamiltonian System

The dynamics of a port-Hamiltonian system is defined as follows.

Definition 2.3. Consider a state space $\mathcal{X}$ and a Hamiltonian

$$H : \mathcal{X} \rightarrow \mathbb{R},$$

defining energy-storage. A port-Hamiltonian system on $\mathcal{X}$ is defined by a Dirac structure

$$\mathcal{D} \subset T_x \mathcal{X} \times T_x^* \mathcal{X} \times \mathcal{F}_R \times \mathcal{E}_R \times \mathcal{F}_P \times \mathcal{E}_P,$$

having energy-storing port $(f_S, e_S) \in T_x \mathcal{X} \times T_x^* \mathcal{X}$ and a resistive structure

$$\mathcal{R} \subset \mathcal{F}_R \times \mathcal{E}_R,$$

corresponding to an energy-dissipating port $(f_R, e_R) \in \mathcal{F}_R \times \mathcal{E}_R$. Its dynamics is specified by

$$\begin{aligned} \left(-\dot{x}(t), \frac{\partial H}{\partial x}(x(t)), f_R(t), e_R(t), f_P(t), e_P(t) \right) &\in \mathcal{D}(x(t)), \\ (f_R(t), e_R(t)) &\in \mathcal{R}(x(t)), \quad t \in \mathbb{R}. \end{aligned} \quad (2.28)$$